Polarization in SEDust: Implementation Notes

Abstract

These are maintenance notes for the polarization capability added to SEDust: where the polarized optics come from, how they are assembled, which implementation choices were made and why, and how far the numbers have been checked. The reader is assumed to be someone extending or repairing this code, not someone calling it. The calling interface is documented in `SEDust_user_manual.pdf`; the underlying physics is documented in `aligned_grain_polarization.pdf`. Neither is repeated here.

Contents

1. Scope, and what lives elsewhere

Three documents cover this work and they do not overlap (Table 1). This one answers only the question *how was this built and why that way*. Where physics is needed to justify a coding choice, the relevant section of the theory report is cited rather than restated.

Question	Document
Why polarization at all; alignment theory; the polarized transfer equation; sign and index conventions; validation targets	`aligned_grain_polarization.pdf` (§1–§5, §9–§11)
How an RT code calls the library: signatures, units, what is already integrated	`SEDust_user_manual.pdf`, §5
How the code is built, why each choice was made, what was measured	this note

Table 1: Division of the polarization documentation.

2. The data source

Everything polarized derives from one file, the orientation-resolved Draine & Hensley (2021) astrodust spheroid table:

```
data/dielectric/q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz
```

(porosity $P = 0.20$, iron fraction $f_{\mathrm{Fe}} = 0$, axial ratio $b/a = 1.400$). Its layout is not self-describing, so it is recorded here (Table 2).

Orientation index. The convention is the trap most likely to bite a future reader, and it is worth stating in full (Table 3); see also §3.6 of the theory report.

3. Derived quantities

For a grain whose symmetry axis lies in the plane of the sky and which is perfectly aligned, the polarization cross section is the difference the grain presents to the two linear polarizations, halved:

$$Q_{\mathrm{pol}} = \tfrac{1}{2}\left[Q(\texttt{jori=3}) - Q(\texttt{jori=2})\right], \tag{1}$$

Property	Value
header	12 lines, skipped
payload order	`((Q(jw,jr,jori), jw=0,1128), jr=0,168), jori=1,3`
repeated	three times: Q_{ext}, then Q_{abs}, then Q_{sca}
one record on disk	the 169 sizes of one (jw, $jori$) pair
records read	3 quantities $\times$ 3 orientations $\times$ 1129
total values	$3 \times 3 \times 1129 \times 169 = 1{,}717{,}209$
wavelength axis	`data/dielectric/DH21_wave`, 1129 nodes
size axis	`data/dielectric/DH21_aeff`, 169 nodes
axis file format	two title lines, then the values free-format

Table 2: Layout of the orientation-resolved Q table. The wavelength and size axes are *not* parsed out of the header; they come from the companion grid files, which list the nodes the table was computed on.

`jori`	geometry	meaning
1	$\mathbf{k} \parallel \hat{a}$	propagation along the symmetry axis
2	$\mathbf{k} \perp \hat{a}$, $\mathbf{E} \parallel \hat{a}$	field along the symmetry axis
3	$\mathbf{k} \perp \hat{a}$, $\mathbf{E} \perp \hat{a}$	field across the symmetry axis

Table 3: Orientation index of the DH21 table. $\hat{a}$ is the spheroid symmetry axis.

formed separately from Q_{ext} and from Q_{abs}, giving $Q_{\mathrm{pol,ext}}$ and $Q_{\mathrm{pol,abs}}$. The random-orientation average of the same table is the trace,

$$Q_{\mathrm{ran}} = \tfrac{1}{3}\,[\,Q_1 + Q_2 + Q_3\,], \tag{2}$$

which is the modified picket fence approximation (MPFA; §3.2 of the theory report). Cross sections follow from

$$C = Q\,\pi a_{\mathrm{eff}}^2, \tag{3}$$

with a_{eff} converted to cm. The alignment efficiency is the Hensley & Draine (2023) Table 1 fit,

$$f_{\mathrm{align}}(a) = \frac{f_{\mathrm{max}}}{1 + (a_{\mathrm{align}}/a)^{\alpha}}, \qquad a_{\mathrm{align}} = 0.0749\,\mu\mathrm{m},\ \ \alpha = 1.80,\ \ f_{\mathrm{max}} = 1.00. \tag{4}$$

4. Module map

Table 4 lists what each file is responsible for. Nothing else in the tree touches polarization.

5. Implementation decisions

This is the part worth reading again in six months.

5.1 The accumulation sites, and the two that hide

`sed_grain_loop` branches on `stoch_method` into four cases, and the polarized channel has to be accumulated wherever the total is. There are eight such statements (Table 5). Missing any one of them produces a polarized spectrum that is silently too low rather than an error.

The lesson. The `'draine'` and `'qm'` branches are OpenMP-parallel and accumulate into a thread-local `Jout_local` that is reduced at the end. The `'heuristic'` branch is serial and accumulates into `Jout` directly. A survey of the accumulation sites conducted by searching for `Jout_local` therefore returns only the parallel branches, and an initial pass over this code identified four sites on exactly that basis. Since `'heuristic'` is the *default* solver, that omission would have made the polarized emission come back identically zero in the default configuration while every other solver looked correct. Any future change that adds a term to the emission sum must be checked against Table 5 branch by branch, not by pattern search.

5.2 Why the `'qm'` rescaling is exact

`qm_solve_grain` does not return a temperature distribution; it returns the emitted spectrum with $C_{\rm abs}$ already folded in. The polarized counterpart is therefore obtained by rescaling:

```
if (Cabs_pop(ii, ir) > 0.0_wp) then
   Jpol_local(ii) = Jpol_local(ii) + &
      wpol * emission_qm(ii) * (Cpol_pop(ii, ir) / Cabs_pop(ii, ir)) / &
      (4.0_wp * PI * lam(ii) * 1.0e-3_wp)
end if
```

This is not an approximation. The temperature distribution $P(T)$ is set by the *heating*, which is governed by $C_{\rm abs}$ and is unchanged by the question of which polarization is being read out. Only the emission readout changes, so

$$C_{\rm pol}\int P(T)B_\lambda(T)\,\mathrm{d}T = \frac{C_{\rm pol}}{C_{\rm abs}}\,C_{\rm abs}\int P(T)B_\lambda(T)\,\mathrm{d}T = \frac{C_{\rm pol}}{C_{\rm abs}}\,\texttt{emission\_qm} \tag{5}$$

holds identically, wavelength by wavelength. Wavelengths with zero $C_{\rm abs}$ emit nothing and are skipped rather than divided by zero.

5.3 The $P(T)$ solver was not touched

No change was made to `calc_P`, `narrow_iterative`, or `qm_solve_grain`. This is deliberate and it is what makes stochastic heating automatically correct for polarization: at every site the same temperature weight multiplies both accumulators, one with $\mathrm{d}n\,C_{\rm abs}$ and the other with $\mathrm{d}n\,f_{\rm align}C_{\rm pol}$. The polarized spectrum is thus the same average over the same $P(T)$, and no separate stochastic treatment of the polarized channel is needed (see §5.7 of the theory report for why this is the physically right structure).

5.4 Mixed orientation-average conventions

This is the one open physical wart, and it is documented rather than hidden.

$C_{\rm abs}$ in SEDust comes from our own T-matrix run, whose random-orientation average is computed exactly. $C_{\rm pol}$ comes from the release table, whose available average is the $1/3$ trace, Eq. (2), i.e. the MPFA. The two are different approximations to the same quantity, and the code mixes them: `build_Cpol` adds the polarized channel and deliberately does *not* recompute $C_{\rm abs}$.
Table 6 gives the measured disagreement, $|Q_{\rm abs}^{\rm trace}/Q_{\rm abs}^{\rm exact} - 1|$, on the shared $(\lambda, a_{\rm eff})$ grid.

Reading the far-infrared maximum. The 3.46% worst case occurs at $a = 4.2\,\mu$m, a size carrying $\mathrm{d}n/n_{\rm H} \sim 2.8 \times 10^{-36}$, so it is numerically irrelevant. Restricting to $a \le 0.5\,\mu$m, which covers 99.999% of the geometric cross section, the worst case drops to 1.98%. At $a \ll \lambda$ both averages approach the same Rayleigh limit, which is why the far infrared is well behaved.

Verdict: deliberately deferred in this release. The mixture is left in place for v1.20, as a decision rather than an oversight. The reasoning is recorded here so that it is not rediscovered later. The mixture is harmless for polarized *emission*, which is a far-infrared phenomenon. It is **not** defensible for polarized *extinction* in the ultraviolet or the optical, and column 8 spans the whole wavelength grid. The `Cpol_ext` output of `dust_extinction` is the same quantity by the same route and carries the same caveat unchanged. Anyone using either shortward of the near infrared must resolve this first. The clean resolution is to recompute $C_{\rm abs}$ on the MPFA convention so that both channels share one average. That changes the existing SEDs, so it breaks their byte-identity with the current reference outputs and has to be done as a separate, deliberately validated step rather than folded into this one. The size of the change is not an argument for leaving it: the correctness argument is that two channels of one calculation should not use two different orientation averages.
What was decided for this release is only *when*, not *whether*. The planned use of these optics — redoing Seon (2018) in the V band, with one or two further optical bands possible — lies entirely in the range where the disagreement is 0.080% or below, while unifying the convention would change $C_{\rm abs}$ and break the byte-identical regression baseline that the rest of v1.20 was validated against. Spending that baseline buys nothing inside the present scope. It has to be spent before any ultraviolet polarized extinction is quoted, and that is the trigger for doing the work.

5.5 $f_{\rm align}$ is the fit, not the release column

`data/release/size_distribution.dat` also ships an alignment column. It agrees with Eq. (4) to a median ratio of 1.0000 but differs by up to 0.32% at the small end. The analytic form is the canonical one in SEDust because it is the published fit, because it is defined on any radius grid rather than only the tabulated one, and because it keeps the alignment model in a single place. Do not mix the two sources: a quantity computed half from one and half from the other is neither.

5.6 Names and surface area

`Cpol` was not renamed to `Cpol_abs`. The name appears at roughly 40 sites and is part of the public surface (`grain_pop_t%Cpol`, the optional `set_pop(..., Cpol_in=...)`). A rename would touch every one of them for no functional gain. The declaration carries an explicit note instead: `Cpol` is the absorption one, `Cpol_ext` its extinction twin.

The polarized extinction was exported as a table column first. A dedicated `dust_cpol_ext` accessor was considered at the time and rejected: `dust_lib` was then an emission-facing API with no extinction surface at all, so such an accessor would have been the module's only extinction entry point, a worse inconsistency than the inconvenience it removed. The condition attached to that decision was that `Cpol_ext` should join a full extinction API if one ever appeared. It has; see §5.7.

5.7 Why the accessor exists now

The table-only export was right for a library that emitted and did not extinguish. Rebuilding a radiative transfer code on top of SEDust changed the premise: the host would have obtained its emission from a call and its opacity from a parsed text file, and that asymmetry is the whole problem. The file is also written on whatever wavelength grid the driver used, so a host on any other grid has to interpolate optics that the model could have evaluated exactly.

`dust_extinction` closes it. The model object already held everything required, so the accessor adds no physics and no new data source; it performs the same size integral the driver performs, over the populations already built. Two fields were added to `grain_pop_t` to make that possible: `Cpol_ext`, previously a module array only, and `gsca`, the scattering asymmetry, which the emission path never needed and which `sed_init` now interpolates onto the size grid beside $C_{\rm abs}$ and $C_{\rm sca}$, from the same Q table and by the same routine the driver used. Agreement with the table is at the precision the file is written with (a median relative deviation of $\sim 7\times10^{-9}$ in each cross section), which is the expected result for one integral evaluated twice.

The table remains, unchanged byte for byte, and remains the right route for a tool that wants the optics without linking the library.

5.8 Compression handling, and a scratch file that leaked

The table ships compressed and Fortran cannot read a deflate stream. Adding a zlib binding was rejected as too heavy for one file. The reader instead shells out once,

```
call execute_command_line('gzip -dc "'//trim(gz_file)//'" > "'// &
                          trim(out_file)//'"', exitstat=estat, cmdstat=cstat)
```

writes `q_jori_scratch.dat` in the working directory, reads it, and deletes it. A fresh clone works with no manual setup step, and neither a 17 MB untracked sibling in `data/dielectric` nor a new build dependency is created. A path not ending in `.gz` is opened directly.

The bug this had. Shell redirection creates the target file *before* `gzip` runs, so when decompression failed the early return left a zero-length `q_jori_scratch.dat` behind in the user's working directory. The failure path now calls `discard_scratch_copy` before bailing out, like every other error exit. The general shape of the bug, an error path that skips the removal the success path performs, is worth watching for in any future edit to this reader: it has eight separate early returns.

6. Verification

6.1 Polarized extinction against the release

The comparison against the release is restricted to $\lambda \geq 0.11\,\mu$m because the reference file turns negative below that. The 0.9422% maximum deviation is not a systematic offset: it sits at $\lambda = 0.112\,\mu$m, right at the sign reversal of the polarized extinction, where the quantity passes through zero and a relative deviation is not a meaningful measure. Away from that point the agreement is at the level of the median.

The 7×10^{-8} figure is the precision the ASCII output format carries, so column 8 and `calc_polext.x` agree as closely as the files allow. They are two independent routes to the same quantity: `calc_polext.x` works straight from `qpol_ext` and the release size distribution, while column 8 goes through `sed_init`, the $\log a$ interpolation onto the SED size grid, and `build_Cpol`.

6.2 Polarized emission

Intrinsic polarization fraction `lamI_pol/lamI_total` under a Mathis ISRF at $U = 1.585$:

The mid-infrared zero is correct rather than a failure: there the emission is dominated by stochastically heated PAHs, which HD23 take to be unaligned.

19.15% versus Planck's 22% is a property of the model, not a bug. Planck measures $p_{\rm max} = 22.0\,(+3.5, -1.4)\%$ at 353 GHz. The 19.15% here is the *maximum* attainable value, obtained before any geometric factor, and every real line of sight reduces it, so the model as implemented cannot reach the observed maximum. This traces to the dust model itself: the axial ratio $b/a = 1.4$ is the minimum non-sphericity DH21b found the starlight polarization efficiency to require, and it was not tuned to maximize the submillimeter polarization fraction. Do not close the gap by adjusting $f_{\rm align}$ or the axial ratio without recognizing that as a change to the dust model, requiring its own revalidation against the extinction and emission constraints HD23 fit.

7. First-principles orientation-resolved optics

The table of §2. is Draine & Hensley's precomputed one. SEDust can now regenerate the same orientation-resolved cross sections from the astrodust dielectric function with its own T-matrix engine, so the polarized optics no longer have to be taken on trust from the release file. The random-orientation optics were already SEDust's own (the `q_astrodust` table read by `q_table`); this closes the remaining gap, the orientation-resolved part read by `q_table_jori`.

Recall the three orientations of the incident wave relative to the spheroid symmetry axis $\hat{a}$: jori = 1 is $\mathbf{k} \parallel \hat{a}$; jori = 2 is $\mathbf{k} \perp \hat{a}$ with $\mathbf{E} \parallel \hat{a}$; jori = 3 is $\mathbf{k} \perp \hat{a}$ with $\mathbf{E} \perp \hat{a}$. The polarized and mean combinations are $Q_{\rm pol} = \frac{1}{2}(Q_3 - Q_2)$ and $Q_{\rm ran} = \frac{1}{3}(Q_1 + Q_2 + Q_3)$.

The optics are reached two ways. The *table* is loaded once and interpolated: this is the production path, fast and safe to call from many threads. The *direct* routine `oriented_cross_sections` computes one node from first principles; it is the shared core the table generator loops over, and it also serves an arbitrary (a, λ) or a shape or porosity for which no table exists. It must not be called from inside a loop over radiative-transfer cells: each call is a full T-matrix solve plus an angular integral, and the Mishchenko routines carry their state in `COMMON` blocks and are not thread-safe.

7.1 Rayleigh regime, $x < 0.1$

In the electrostatic limit the oblate spheroid has a diagonal polarizability $\boldsymbol{\alpha} = \mathrm{diag}(\alpha_a, \alpha_b, \alpha_b)$, with α_a for $\mathbf{E}$ along the symmetry axis and α_b for $\mathbf{E}$ transverse, set by the depolarization factors L_a, L_b of the spheroid through $\alpha_i \propto (\varepsilon - 1)/[1 + (\varepsilon - 1)L_i]$. Absorption and scattering follow as $Q_{\mathrm{abs}} \propto \mathrm{Im}\,\alpha$ and $Q_{\mathrm{sca}} \propto |\alpha|^2$. The three orientations sample exactly these two polarizabilities: jori = 1 and 3 present the transverse α_b (the field is transverse to $\hat{a}$ in both), jori = 2 presents α_a. So the orientation-resolved cross sections are the two polarization components the random average $(\alpha_a + 2\alpha_b)/3$ is already built from; extracting them before the average is analytic and exact, and Q(jori = 1) equals Q(jori = 3) identically. `rayleigh_limit` returns them through optional arguments, leaving its averaged outputs untouched.

7.2 T-matrix regime, $0.1 \le x \le 50$

The T-matrix $\mathbf{T}$ is a property of the particle alone — size, shape, refractive index — independent of orientation and of the incidence and scattering directions. SEDust solves it once for each wavelength and radius with Mishchenko's `tmd_one_scatmat`. Two orientation-resolved quantities are then read off the same $\mathbf{T}$.

Extinction, from the optical theorem. Mishchenko's `AMPL` evaluates the fixed-orientation amplitude matrix $\mathbf{S}(\mathbf{k} \to \mathbf{k}')$ from the stored $\mathbf{T}$ for a particle placed at chosen Euler angles. Taking the scattering direction equal to the incidence direction (forward) and applying the optical theorem, $C_{\mathrm{ext}} = (4\pi/k)\,\mathrm{Im}\,S_{\mathrm{fwd}}$ for the incident polarization, gives C_{ext} for each orientation: the symmetry axis is placed along $\mathbf{k}$ (jori = 1) or across it (jori = $2, 3$) and the co-polar forward amplitude is read ($S_{\theta\theta}$ for $\mathbf{E}$ along $\hat{\theta}$, $S_{\phi\phi}$ for $\hat{\phi}$).

Scattering, from the phase-matrix integral. The same `AMPL`, swept over all scattering directions at fixed incidence, gives the differential scattering cross section $|\mathbf{S}|^2$; the scattering cross section is its integral over the sphere, $C_{\mathrm{sca}} = \int (|S_{\theta \leftarrow p}|^2 + |S_{\phi \leftarrow p}|^2)\, d\Omega$ for incident polarization p, and absorption follows by conservation, $C_{\mathrm{abs}} = C_{\mathrm{ext}} - C_{\mathrm{sca}}$. The fixed-orientation scattered intensity is a trigonometric polynomial of degree $\sim 2N_{\mathrm{max}}$ in $\cos\theta$ and in ϕ, so Gauss–Legendre nodes in $\cos\theta$ ($N_{\mathrm{max}} + 2$ of them) and a uniform azimuth grid ($2N_{\mathrm{max}} + 2$ points) integrate it to machine precision.

An engine subtlety. Mishchenko's random-orientation expansion `GSP` reuses the `/TMAT/` storage as scratch through an `EQUIVALENCE` and so destroys the T-matrix it reads. The amplitude evaluation needs that same T-matrix afterward, so `tmd_one_scatmat` keeps an intact copy in a second common block and restores `/TMAT/` after `GSP`. Without this the oriented cross sections came out as garbage (negative C_{ext}). This corrects a premise stated in the planning note, that the converged T-matrix simply remains available after the solve.

7.3 Geometric-optics regime, $x > 50$

Here the T-matrix does not converge and an asymptotic model is used. Extinction is the extinction paradox, $C_{\rm ext} \to 2A_{\rm proj}$, twice the geometric shadow. For an oblate spheroid of equatorial semi-axis a_s and symmetry semi-axis c ($a_s/c = b/a$), the shadow is πa_s^2 when **k** is along the symmetry axis (jori = 1) and $\pi a_s c$ when it is across (jori = 2, 3), giving $Q_{\rm ext}(1) = 2(b/a)^{2/3}$ and $Q_{\rm ext}(2) = Q_{\rm ext}(3) = 2(b/a)^{-1/3}$. The shadow depends on the direction of **k** relative to the axis but not on the incident polarization, so extinction carries no polarization at this order.

The polarization in this regime is carried by absorption. In the opaque limit, where a refracted ray is fully absorbed — valid when $\mathrm{Im}(m)\,x \gg 1$ — the absorptivity of an illuminated surface element is one minus its Fresnel power reflectance, and that reflectance is polarization-dependent ($R_s \neq R_p$ away from normal incidence). Integrating over the illuminated half of the spheroid,

$$C_{{\rm abs},p} = \int_{\rm illum} \left[1 - \left(|\hat{e}\cdot\hat{s}|^2 R_s(\theta_i) + |\hat{e}\cdot\hat{p}|^2 R_p(\theta_i)\right)\right] dA_{\rm proj},$$

where θ_i is the local angle of incidence and $\hat{s}, \hat{p}$ the local senkrecht and parallel directions. The two grain-frame polarizations ($\hat{e} = \hat{a}$ versus $\hat{e} \perp \hat{a}$) project differently onto the local $\hat{s}, \hat{p}$ basis as they sweep the curved surface, and that is the whole source of the dichroism. A purely geometric ray treatment, with no Fresnel coefficients, would give no polarization at all; the surface reflectance is what makes the regime dichroic, and it is needed to match the release, which carries a real absorption dichroism there — about 3% of the mean, and *growing* with size parameter, the wave-optics signature a ray limit cannot reproduce. This is a deliberate approximation, documented at the code site with its opaque-limit domain of validity; it would matter for a size distribution with large-grain weight, which the astrodust distribution does not have.

7.4 Dispatch, generation, and drop-in

`oriented_cross_sections` selects the regime by x and returns the oriented ($Q_{\rm ext}, Q_{\rm abs}, Q_{\rm sca}$) for one node, with the same branch boundaries and fallbacks as the random-orientation driver. `run_q_jori.x` loops it over the DH21 grid and writes the table in the exact stream the release reader expects (§2.), so the result is a drop-in: `sed_init` and `build_astrodust` already take an optional `qpol_path` that defaults to the release file, and pointing it at the regenerated table switches the polarized optics to SEDust's own with no code change. A full sweep is dominated by the T-matrix nodes (of order 16 hours on one core), so the driver has a wavelength-window `range` mode for process-level parallelism and a `merge` mode that reassembles the windows — a window is not a contiguous slice of the stream, because the wavelength index runs innermost.

7.5 How far it has been checked

Each regime was compared node by node against the release table (Table 9). Across the Rayleigh and T-matrix regimes — every wavelength and radius that carries astrodust weight — the reconstruction matches the release to a median of a few parts in 10^4, in both the mean and the polarized cross sections. The geometric-optics region agrees only coarsely, as an asymptotic model should: the mean absorption to about ten percent and the dichroism to about three-quarters of the release value, both with the correct sign and size-parameter trend. It joins the T-matrix continuously across $x = 50$ in sign and ordering, and it carries negligible weight in the size integral.

End to end, feeding the regenerated table through `calc_polext` reproduces the released polarized extinction to a median of 0.06% over $\lambda \geq 0.11\,\mu$m, against 0.03% for the release table itself; the two agree with each other to a median of 0.02% at the level of the observable. So the few-parts-in-10^4 reconstruction error at the level of a single cross section survives the size integral as a few parts in 10^4 in the polarized extinction, computed entirely from the astrodust dielectric function with no recourse to the release optics.

8. Limits and extension paths

These are limits of the published optics, not of the code.

Birefringence is now computed; the release table has none. The release table stores $Q_{\rm ext}$, $Q_{\rm abs}$ and $Q_{\rm sca}$ only, no phase retardation, so from it $C_{\rm circ} = 0$. The first-principles route of §7. already evaluates the fixed-orientation amplitude matrix (Mishchenko's `AMPL`, in `tmatrix/src/ampl_oriented.f`) and used only its imaginary forward part, for extinction. Keeping the real part as well costs nothing — it comes from the same amplitude call — and gives the birefringence, the phase retardation that couples U and V. It is stored as an optional 4th block of the regenerated table, $Q_{\rm re}({\rm jori}) = (4\pi/k)\,{\rm Re}[S_{\rm fwd}]/(\pi a^2)$, the real twin of $Q_{\rm ext}$, from which $C_{\rm bir} = \frac{1}{2}(Q_{\rm re,3} - Q_{\rm re,2})\,\pi a^2$. The reader exposes it as `qbir_ext` with a `has_bir` flag that stays `.false.` for an older 3-block table, so nothing about the default path changes. No astrodust reference for the birefringence exists, so it is certified internally by Kramers-Kronig: Re and Im of the forward-amplitude difference are a Hilbert pair, and the directly computed birefringence agrees with the Hilbert transform of the already-validated dichroism to a median $\sim 0.1\%$ over the interior of the grid (the transform itself self-tests to 2×10^{-5} on a Lorentz oscillator; the residual is the finite-range truncation). What remains is to return it through `dust_extinction` and to consume it in the $U \leftrightarrow V$ transfer term, both on the host side.

No scattering matrix for aligned grains. The release gives the integrated $Q_{\rm sca}$ with no angular information, so scattering by an aligned spheroid cannot be modeled from it. Reaching optical polarized transfer would require wavelength-resolved Mueller matrices, from `ampld.lp.f`, DDSCAT, ADDA or CosTuuM, plus a matrix-based scattering treatment on the consumer side. The data volume is $\lambda \times \theta \times$ orientation $\times$ 16 components, which is the reason this has not been attempted.

The random-orientation scattering matrix does exist, in five optical bands. This is a different object from the previous paragraph and should not be confused with it. Dropping the alignment, a randomly oriented spheroid with a plane of symmetry has six independent scattering-matrix elements rather than sixteen, and those *are* computed in the tree: `tmatrix/driver/run_scatmat.f90` restores Mishchenko's `MATR` expansion, accumulates the single-size matrices over the HD23 size distribution with $C_{\rm sca}\,{\rm d}n/n_{\rm H}$ weighting, and writes `tmatrix/output/scatmat_astrodust_P0.20_Fe0.00_1.400.dat`: one block per wavelength, 181 rows at $\theta = 0, 1, \ldots, 180^\circ$, with columns $F_{11}, F_{22}, F_{33}, F_{44}, F_{12}, F_{34}$. It answers what an unaligned grain population scatters; it does not answer the aligned question above.

The stored wavelengths are 0.36, 0.44, 0.55, 0.64 and 0.79 μm, roughly *UBVRI*. That set was chosen, and the full sweep deliberately not run, because the planned use is the V band first with one or two further optical bands possible, and neither the far infrared nor the ultraviolet is in scope. The whole 1129-point grid costs about 1.7 hours and would be spent almost entirely on wavelengths nothing is waiting for. Extending is cheap and needs no code change: `./run_scatmat.x all` does the full grid, or a list of wavelengths does a subset, at about 5.5 s per wavelength.

Far infrared needs neither. In that regime the albedo is essentially zero, so the transfer requires only the extinction matrix and the emission vector, both fully determined by what is exported. The HAWC+ 154 μm comparison lies inside this regime, so the far-infrared milestone is reachable with the present data and should be attempted before any scattering work.

Astrodust only. DL07 and Zubko have no polarized optics. `build_dl07` releases any polarized arrays left by a previous astrodust initialization, so `lamI_pol` returns zeros rather than stale values.

8.1 Cost of `build_Cpol`

Loading the orientation-resolved table is the most expensive part of model construction (Table 10).

The orientation-resolved arrays `qext_j`, `qabs_j` and `qsca_j` are deallocated as soon as the combinations of Eqs. (1)–(2) have been formed. They are intermediate quantities: nothing downstream of `build_Cpol` reads them, and every later step uses the five derived arrays only. Releasing them removes 13.7 MB, about two-thirds of what the table would otherwise hold, and leaves 7.6 MB resident for the rest of the run. The three arrays are `private` to `q_table_jori`, so no consumer can depend on their surviving the build.

Measured on `main_astrodust.x`, the maximum resident set size falls from 51232 kB to 40088 kB, a reduction of about 10.9 MB. The peak does not fall by the full 13.7 MB because it is a high-water mark: once the orientation-resolved arrays are gone, the later SED solution stages set the peak instead. The steady-state occupancy just after `sed_init` does drop by the full 13.7 MB.

9. Reproducing the numbers

From `SEDust/sed/`:

```
make calc_polext.x # standalone polarized-extinction check
./calc_polext.x # -> output/polarized_extinction_ours.dat
                          # lambda, ours, reference, ratio

make calc_kext_astrodust.x # full-pipeline extinction table
./calc_kext_astrodust.x # -> ../data/kext_astrodust_MW.dat, 8 columns

make libsedust.a # library, for the emission path
```

`calc_polext.x` prints the median and maximum deviation against the release directly, restricted to $\lambda \geq 0.11\,\mu$m. Column 8 of the extinction table should reproduce it to the ASCII precision quoted in Table 7; if the two disagree by more than that, the $\log a$ interpolation in `build_Cpol` or the size grid it interpolates onto is the place to look, since that is the only step separating the two routes.

For the emission fractions of Table 8, build a model with `build_astrodust`, evaluate `J_Mathis` at $U = 1.585$, and call `dust_emission` with `lamI_pol` requested; the example in §5.4 of the user manual is the complete recipe.

10. References

- Draine, B. T., & Hensley, B. S. 2021b, ApJ, 919, 65 (DH21b). `2021ApJ...919...65D`
- Hensley, B. S., & Draine, B. T. 2023, ApJ, 948, 55 (HD23). `2023ApJ...948...55H`
- Peest, C., et al. 2023, A&A, 673, A112. `2023A&A...673A.112P`
- Planck Collaboration 2020, A&A, 641, A12. `2020A&A...641A..12P`
- Reissl, S., Wolf, S., & Brauer, R. 2016, A&A, 593, A87 (POLARIS I). `2016A&A...593A..87R`
- Seon, K.-I. 2018, ApJ, 862, 87. `2018ApJ...862...87S`

File / entity	Responsibility
`sed/src/q_table_jori.f90`	The reader. Decompresses, parses the three quantity blocks, validates (monotone axes, finite values, no trailing data), forms Eqs. (1)–(2), and supplies `falign_hd23(a)`, Eq. (4). Publishes `qpol_ext`, `qpol_abs` and `qran_*`; the orientation-resolved `qext_j`/`qabs_j`/`qsca_j` are `private` and are deallocated once the combinations have been formed.
`sed_astrodust.f90`: `build_Cpol`	Interpolates $Q_{\rm pol}$ from the 169-node table axis onto the size-distribution grid in $\log a$, exactly as $C_{\rm abs}$ and $C_{\rm sca}$ are, multiplies by $\pi a_{\rm eff}^2$, and fills `falign_ad`. Checks that the polarized and Q wavelength grids agree node for node.
`sed_astrodust.f90`: `Cpol`, `Cpol_ext`, `falign_ad`	Module-level arrays, `(NLAM,NA)` and `(NA)`. `Cpol` is the *absorption* one and drives polarized emission; `Cpol_ext` is its extinction twin and drives dichroic extinction.
`dust_model_mod.f90`: `grain_pop_t`	Carries `Cpol(:,:)` and `falign(:)`. Both unallocated is how the solver recognizes an unpolarized population. Also carries the extinction-side `Cpol_ext(:,:)` and `gsca(:,:)`, read only by `dust_extinction`.
`sed_astrodust.f90`: `set_pop`	Optional `Cpol_in`, `falign_in`. Both must be supplied together; supplying neither leaves the population unpolarized. The extinction-side `Csca_in`, `Cpol_ext_in` and `gsca_in` are optional and independent; a population that omits them contributes zero to those terms of the size integral.
`sed_astrodust.f90`: `sed_grain_loop`	Optional `Cpol_pop`, `falign_pop`, `Jpol`, again all three or none (`do_pol`). Accumulates the polarized channel alongside `Jout` in every solver branch (§5.1).
`sed_astrodust.f90`: `dust_emission`	Optional `lamI_pol`, last argument. Sums `Jpol` over the populations that carry polarized optics and applies the same unit convention as `lamI_total`.
`sed_astrodust.f90`: `dust_extinction`	The extinction accessor. Integrates $C_{\rm abs}$, $C_{\rm sca}$, the scattering-weighted $\langle\cos\theta\rangle$ and $C_{\rm pol,ext}\,f_{\rm align}$ over the size distribution of every population, and returns them per H atom on `m%lam`. These are the same numbers column 8 of the table carries, without the file (§5.7).
`sed/src/calc_polext.f90`	Standalone check: computes the polarized extinction directly from `qpol_ext` and the release size distribution, and compares against the published `polarized_extinction.dat`. Independent of `sed_init`.
`sed/src/calc_kext_astrodust.f90`	Writes column 8, `C_polext/H`, of `data/kext_astrodust_MW.dat` as $\sum_a \mathrm{d}n_{\rm Ad}(a)\,C_{\rm pol,ext}(\lambda,a)\,f_{\rm align}(a)$.

Table 4: Responsibility of each file touched by the polarization update.

branch	context	accumulates into	pattern
`'draine'`	equilibrium grain	`Jpol_local`	thread-local
`'draine'`	stochastic, sum over $P(T)$	`Jpol_local`	thread-local
`'heuristic'`	equilibrium grain	`Jpol`	**direct**
`'heuristic'`	stochastic, sum over $P(T)$	`Jpol`	**direct**
`'qm'`	QM solve, rescaled	`Jpol_local`	thread-local
`'qm'`	GD fallback after QM failure	`Jpol_local`	thread-local
`'qm'`	equilibrium grain	`Jpol_local`	thread-local
`'equil'`	every grain	`Jpol`	direct

Table 5: Where the polarized emission is accumulated. The two boldface rows are the ones a search for the `Jout_local` pattern does not find.

band	median	maximum
far infrared ($\lambda > 30\,\mu$m)	0.022%	3.46%
restricted to $a \leq 0.5\,\mu$m		1.98%
mid infrared	0.033%	
near infrared	0.038%	
optical	0.080%	
ultraviolet	0.21%	9.53%

Table 6: Disagreement between the trace average and the exact orientation average of $Q_{\rm abs}$.

Check	Result
Column 8 vs. released `polarized_extinction.dat`	median 0.0296%, max 0.9422%, 992 points
Column 8 vs. `calc_polext.x`	agree to 7×10^{-8} relative
Columns 1–7 of `kext_astrodust_MW.dat`	byte-identical to the pre-change file
`astrodust_irem_ours_S1/S2/PAH.dat`	byte-identical
`dl07_sed_ours_mw31_60.dat`	byte-identical
Build	zero warnings

Table 7: Verification of the polarized extinction path and regression of everything else.

λ	fraction
12 μm	0.00%
100 μm	13.67%
154 μm	17.15%
350 μm	18.79%
850 μm	19.15%
median over 300–3000 μm	19.18%

Table 8: Intrinsic polarization fraction of the emitted spectrum, before any geometric factor.

Regime	x	Quantity	median $\lvert$ours/HD23 $-$ 1$\rvert$
Rayleigh	< 0.1	$Q_{\rm ran}$ (abs)	2.3×10^{-4}
		$Q_{\rm pol}$ (abs)	3.7×10^{-4}
T-matrix	0.1–50	$Q_{\rm ran}$ (ext)	2.1×10^{-4}
		$Q_{\rm pol}$ (ext)	8.4×10^{-4}
		$Q_{\rm ran}$ (abs)	1.9×10^{-4}
		$Q_{\rm pol}$ (abs)	7.6×10^{-4}
GO	> 50	$Q_{\rm ran}$ (abs)	9.3×10^{-2}
		$Q_{\rm pol}$ (abs)	3.3×10^{-1}

Table 9: Node-by-node reconstruction against the HD23 release table. The geometric-optics dichroism is at ~ 0.75 of the release with matching sign and trend; its size-parameter region carries no astrodust weight.

Item	Size
compressed file	4.1 MB
scratch file when expanded	17.2 MB
raw arrays `qext_j`, `qabs_j`, `qsca_j`	13.7 MB, transient
derived `qpol_*`, `qran_*`	7.6 MB, kept
resident after `sed_init`	7.6 MB
time	about 1 s of the 3.1 s model build

Table 10: Cost of loading the orientation-resolved table. The orientation-resolved arrays exist only between the read and the formation of the derived combinations.